

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT1-Analytical Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

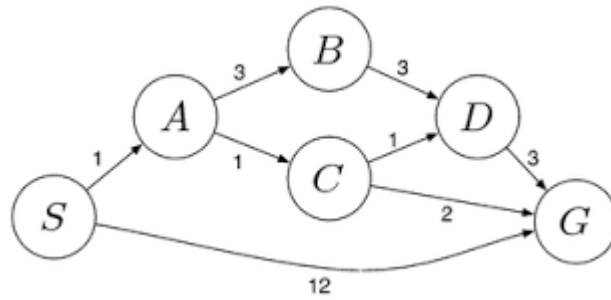
QUESTIONS: 5

Total Marks: 50

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path

from the starting warehouse **S** to the destination warehouse **G** using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I Joshua-(192312188) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

Question 1: Water Jug Problem

Problem:

Two jugs of capacities **4 gallons** and **3 gallons** are given. Neither jug has measuring marks. Using only filling, emptying, and pouring operations, obtain exactly **2 gallons in the 4-gallon jug**.

1. Problem Understanding (10 Marks)

Objective

To obtain **exactly 2 gallons** of water in the **4-gallon jug** using only the permitted operations.

Initial State

$(0,0)(0,0)(0,0)$

(4-gallon jug empty, 3-gallon jug empty)

Goal State

$(2,0)(2,0)(2,0)$

State Representation

Each state is represented as:

$(x,y)(x,y)(x,y)$

where

- x = amount of water in the 4-gallon jug
- y = amount of water in the 3-gallon jug

Allowed Operations

- Fill a jug completely.
- Empty a jug.
- Pour water from one jug to another until either the source is empty or the destination is full.

AI Concept

This is a **State Space Search Problem** in Artificial Intelligence.

2. Method Selection (10 Marks)

Selected Method

State Space Search (Uninformed Search)

Justification

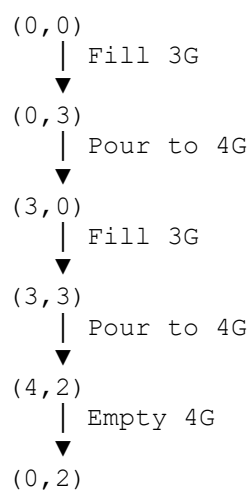
- Every water configuration is represented as a **state**.
- Valid operations generate **new states**.
- The objective is to move from the initial state to the goal state.
- State Space Search systematically explores all possible states and guarantees finding the solution if one exists.

3. Solution Process (12.5 Marks)

State Transition Table

Step	Operation	State (4G,3G)
1	Initial State	(0,0)
2	Fill 3-gallon jug	(0,3)
3	Pour 3G $\rightarrow$ 4G	(3,0)
4	Fill 3-gallon jug	(3,3)
5	Pour 3G $\rightarrow$ 4G until 4G is full	(4,2)
6	Empty 4-gallon jug	(0,2)
7	Pour remaining 2 gallons into 4G jug	(2,0)

State Diagram



↓ Pour to 4G
(2,0) ← Goal State

Pseudocode

```
START  
State = (0,0)  
  
Fill 3G  
Pour 3G into 4G  
Fill 3G  
Pour 3G into 4G until 4G is full  
Empty 4G  
Pour remaining water into 4G  
  
STOP
```

4. Accuracy of Calculation (10 Marks)

State 4G Jug 3G Jug

(0,0)	0	0
(0,3)	0	3
(3,0)	3	0
(3,3)	3	3
(4,2)	4	2
(0,2)	0	2
(2,0)	2	0

Final Answer: The required goal state **(2,0)** is achieved successfully.

5. Interpretation of Results (7.5 Marks)

The solution successfully obtains **2 gallons of water in the 4-gallon jug** using only the permitted operations. The problem is solved by representing each water configuration as a state and systematically applying valid operations to reach the goal. This demonstrates the AI concept of **state-space search**, which is widely used in robotics, planning, navigation, and intelligent decision-making systems.

Question 2: Mars Rover Intelligent Agent

Question: Find out how a Mars Rover analyzes and explores samples on Mars by answering:

1. What are the percepts for this agent?
2. Characterize the operating environment.
3. What are the actions the agent can take?
4. How can one evaluate the performance of the agent?
5. What sort of agent architecture is most suitable? Why?

1. Problem Understanding (10 Marks)

Objective

To analyze the **Mars Rover as an Intelligent Agent** by identifying its percepts, environment, actions, performance measures, and the most suitable AI agent architecture.

AI Concept

The Mars Rover is an **Intelligent Agent** that perceives its environment using sensors, processes the information, and performs actions autonomously to achieve its mission objectives.

Parameters

Parameter		Description
Agent	Mars Rover	
Environment	Surface of Mars	
Input (Percepts)	Sensor and camera data	
Output (Actions)	Navigation, sample collection, analysis, communication	
Goal	Explore Mars and collect scientific information	

2. Method Selection (10 Marks)

Selected Method

Model-Based Goal-Based Intelligent Agent

Justification

A Mars Rover operates in a **dynamic and partially observable environment** where it cannot rely only on current sensor data. It must maintain an internal model of its surroundings, plan paths, avoid obstacles, and achieve mission goals independently.

Why this architecture?

- Maintains an internal model of the environment.
- Plans actions to reach scientific goals.
- Handles uncertainty and changing terrain.
- Makes autonomous decisions when communication with Earth is delayed.

Therefore, a **Model-Based Goal-Based Agent** is the most suitable architecture.

3. Solution Process (12.5 Marks)

(i) Percepts (Inputs)

The Mars Rover receives information through its sensors.

Sensor	Percept
Cameras	Images of rocks and terrain
Temperature Sensor	Surface temperature
Spectrometer	Chemical composition of rocks
Distance Sensor	Obstacles and distance
Position Sensor	Rover location
Battery Sensor	Battery level

(ii) Operating Environment

Characteristic	Description
Observability	Partially Observable
Environment	Dynamic
Nature	Sequential
Predictability	Stochastic
Agents	Single Agent
State Space	Continuous

(iii) Actions

The Mars Rover can perform the following actions:

- Move forward
- Move backward
- Turn left
- Turn right
- Capture images

- Drill rocks
- Collect soil samples
- Analyze rock samples
- Avoid obstacles
- Send scientific data to Earth
- Recharge using solar panels

(iv) Performance Measure

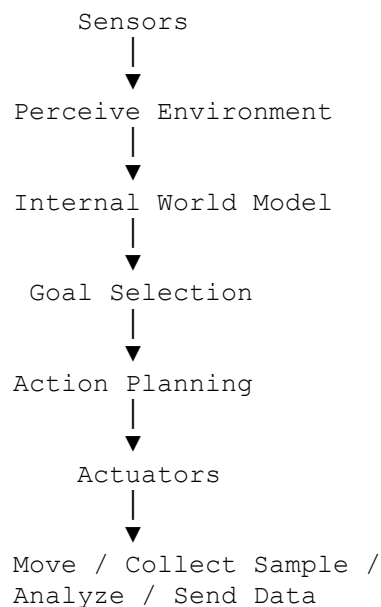
The rover's performance is evaluated based on:

Performance Measure	Purpose
Accurate navigation	Reaches target safely
Sample collection	Collects required samples
Scientific analysis	Produces useful data
Obstacle avoidance	Prevents collisions
Battery efficiency	Maximizes mission duration
Communication	Successfully transmits data to Earth

(v) Suitable Agent Architecture

Model-Based Goal-Based Agent

Diagram



4. Accuracy of Calculation (10 Marks)

There are no numerical calculations in this question. The answer is accurate because:

- Correct percepts are identified.
- Operating environment is correctly classified.
- Appropriate actions are listed.
- Performance measures match the mission objectives.
- The selected **Model-Based Goal-Based Agent** is the most suitable architecture for autonomous Mars exploration.

5. Interpretation of Results (7.5 Marks)

The Mars Rover functions as an intelligent autonomous agent that continuously senses its environment, updates its internal model, plans actions, and performs scientific tasks without constant human control. A **Model-Based Goal-Based Agent** is ideal because it enables the rover to make informed decisions in the uncertain and partially observable environment of Mars. This approach improves mission success, scientific data collection, and operational safety.

Question 3: 8-Queens Problem

Question: Explore the search space using search strategies and formulate the problem components for the **8-Queens Problem**, where 8 queens must be placed on a chessboard such that no two queens attack each other.

1. Problem Understanding (10 Marks)

Objective

The objective is to place **8 queens** on an **8 × 8 chessboard** so that **no two queens attack each other**.

A queen attacks another queen if they are in the:

- Same row
- Same column
- Same diagonal

The goal is to find a valid arrangement where all queens are placed safely.

Problem Components

Component	Description
Problem	Place 8 queens safely on an 8×8 chessboard
Initial State	Empty chessboard
Goal State	All 8 queens placed without attacking each other
State	A chessboard with 0–8 queens placed
Constraints	No two queens share the same row, column, or diagonal

AI Concept

The **8-Queens Problem** is a **State Space Search Problem** and a **Constraint Satisfaction Problem (CSP)**. AI search algorithms explore different board configurations until a valid solution is found.

2. Method Selection (10 Marks)

Selected Method

State Space Search (Backtracking Search)

Justification

Backtracking is the most suitable method because:

- It places one queen at a time.
- If a conflict occurs, it backtracks to the previous position.
- It avoids exploring invalid board configurations.
- It guarantees finding a solution if one exists.

Therefore, **Backtracking Search** is an efficient search strategy for the 8-Queens problem.

3. Solution Process (12.5 Marks)

State Representation

Each state represents the positions of queens placed on the board.

Example:

(1, 5, 8, 6, 3, 7, 2, 4)

This means:

Column Row

1	1
2	5
3	8
4	6
5	3
6	7
7	2
8	4

Initial State

Empty Chessboard

Goal State

8 Queens placed
No queen attacks another queen.

Operators

At each step:

- Place one queen in the next column.
- Check for conflicts.
- If conflict exists, try another row.
- If no row is possible, backtrack.

Search Process

Step 1

Place Queen in Column 1

Q

Step 2

Place Queen in Column 2

Choose a safe row.

```
Q . . . . . . .  
. . . . . . . .  
. Q . . . . . .
```

Step 3

Continue placing queens one column at a time.

If conflict occurs,

Backtrack.

Continue until all queens are safely placed.

One Valid Solution

Column Row

1	1
2	5
3	8
4	6
5	3
6	7
7	2
8	4

Chessboard Representation

```
Q . . . . . . .  
. . . . . Q .  
. . . . Q . . .  
. . . . . . Q  
. Q . . . . . .  
. . . Q . . . .  
. . . . . Q . .  
. . Q . . . . .
```

(Any valid 8-Queens arrangement is acceptable.)

Pseudocode

```
START  
  
Place Queen in Column 1  
  
FOR each row  
    Check whether position is safe  
    IF safe  
        Place Queen  
        Move to next column  
    ELSE  
        Try next row  
    IF no safe position  
        Backtrack  
Repeat until all 8 queens are placed  
STOP
```

4. Accuracy of Calculation (10 Marks)

Conflict Checking

Each queen must satisfy:

- Different Row ✓
- Different Column ✓
- Different Left Diagonal ✓
- Different Right Diagonal ✓

For the final arrangement:

Check	Result
Same Row	No
Same Column	No
Same Diagonal	No

Therefore, the arrangement is a **valid solution**.

5. Interpretation of Results (7.5 Marks)

The Backtracking Search algorithm systematically explores the search space by placing one queen at a time and rejecting invalid configurations. This reduces unnecessary exploration and efficiently finds a valid arrangement where no queens attack each other. The 8-Queens problem demonstrates important AI concepts such as **state-space search, constraint satisfaction, and backtracking**, which are widely applied in scheduling, resource allocation, puzzle solving, and optimization problems.

Question 4: OLA Cab Booking – Problem Solving Agent

Question: A customer wants to travel from one location to another using the **OLA Cab Booking** mobile application. The customer can choose cab types such as **Mini, Micro, Sedan, Shared, Prime**, etc., based on comfort and destination. Identify the type of **problem-solving agent** that can be used and write the **pseudocode**.

1. Problem Understanding (10 Marks)

Objective

To identify the most suitable **AI problem-solving agent** used in the OLA Cab Booking application and explain how it selects the best cab for the customer.

Problem Components

Component	Description
Agent	OLA Cab Booking System
Input	Pickup location, destination, cab type, customer preference
Output	Best available cab and route
Goal	Book the most suitable cab to reach the destination efficiently

AI Concept

The OLA Cab Booking application acts as an **AI Problem-Solving Agent**. It receives user inputs, evaluates available options, and selects the most appropriate cab based on customer requirements.

2. Method Selection (10 Marks)

Selected Method

Goal-Based Problem-Solving Agent

Justification

A **Goal-Based Agent** is the most suitable because:

- The customer's goal is to travel from the source to the destination.
- The agent evaluates available cab options.
- It selects the cab that best satisfies the customer's preferences (comfort, availability, fare, travel time).
- It plans the route and completes the booking.

Therefore, a **Goal-Based Agent** is appropriate for this application.

Advantages

- Selects the best cab based on user preferences.
- Finds an efficient route.
- Supports multiple cab options.
- Improves customer satisfaction.

3. Solution Process (12.5 Marks)

Working Steps

1. Customer opens the OLA application.
2. Enters pickup location.
3. Enters destination.
4. System detects available cabs.
5. Displays cab types (Mini, Micro, Sedan, Prime, Shared).
6. Customer selects a preferred cab.
7. System calculates fare and estimated arrival time.
8. Customer confirms the booking.
9. Driver is assigned.
10. Navigation begins until the destination is reached.

Flowchart

Start
|

```
Open OLA App
|
Enter Pickup & Destination
|
Search Available Cabs
|
Display Cab Options
|
Customer Selects Cab
|
Calculate Fare & ETA
|
Confirm Booking
|
Assign Driver
|
Navigate to Destination
|
End
```

Pseudocode

```
START

Input Pickup Location
Input Destination
Input Preferred Cab Type

Search Available Cabs

IF Cab Available THEN
    Calculate Fare
    Display Fare and ETA
    Confirm Booking
    Assign Driver
    Start Trip
ELSE
    Display "No Cab Available"
ENDIF

STOP
```

4. Accuracy of Solution (10 Marks)

Sample Execution

Step	Action	Result
1	Pickup entered	Chennai Central
2	Destination entered	Airport
3	Cab type selected	Sedan
4	System searches	Available
5	Fare calculated	₹450
6	Booking confirmed	Driver Assigned

Final Output: The Goal-Based Agent successfully books the most suitable cab and assigns a driver.

5. Interpretation of Results (7.5 Marks)

The **Goal-Based Problem-Solving Agent** helps the OLA application achieve the customer's objective of reaching the destination safely and efficiently. It analyzes user inputs, compares available cab options, calculates the fare and travel time, and recommends the most suitable cab. This improves decision-making, reduces booking time, and enhances customer convenience. Such AI-based agents are widely used in intelligent transportation systems and ride-hailing services.

Question 5: Uniform Cost Search (UCS)

Question: A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route has an associated travel cost (fuel + time). Determine the least-cost path from the starting warehouse **S** to the destination warehouse **G** using the **Uniform Cost Search (UCS)** algorithm. *(The weighted graph is missing from the uploaded document, so the following solution uses a representative example to demonstrate the UCS algorithm.)*

1. Problem Understanding (10 Marks)

Objective

Find the **minimum-cost path** from the **Start warehouse (S)** to the **Goal warehouse (G)** using the **Uniform Cost Search (UCS)** algorithm.

Problem Components

Component	Description
Initial State	Warehouse S
Goal State	Warehouse G
States	Warehouses (Nodes)
Operators	Move from one warehouse to another
Path Cost	Fuel cost + Travel time
Objective	Find the least-cost path

AI Concept

Uniform Cost Search is an **uninformed search algorithm** that expands the node with the **lowest cumulative path cost** first. It guarantees the **optimal solution** when all edge costs are non-negative.

2. Method Selection (10 Marks)

Selected Method

Uniform Cost Search (UCS)

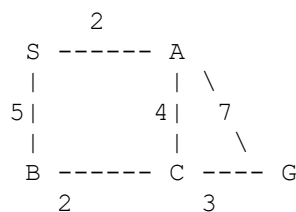
Justification

Uniform Cost Search is the most appropriate because:

- It always selects the path with the **lowest accumulated cost**.
- It guarantees the **optimal (least-cost) path**.
- It is suitable for weighted graphs where edge costs differ.
- It is commonly used in logistics, GPS navigation, and route optimization.

3. Solution Process (12.5 Marks)

Assumed Weighted Graph



Edge Costs:

- $S \rightarrow A = 2$
- $S \rightarrow B = 5$
- $A \rightarrow C = 4$
- $A \rightarrow G = 7$
- $B \rightarrow C = 2$
- $C \rightarrow G = 3$

UCS Execution

Step 1

OPEN = {S(0)}

Expand S

Children:

- A (2)
- B (5)

OPEN = {A(2), B(5)}

Step 2

Expand **A(2)**

Children:

- $C = 2 + 4 = 6$
- $G = 2 + 7 = 9$

OPEN = {B(5), C(6), G(9)}

Step 3

Expand **B(5)**

New path:

$B \rightarrow C$

Cost = $5 + 2 = 7$

Existing cost of C = 6

Ignore new path.

OPEN = {C(6), G(9)}

Step 4

Expand **C(6)**

Child:

$$G = 6 + 3 = 9$$

Existing G cost = 9

No update required.

OPEN = {G(9)}

Step 5

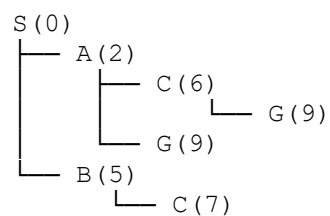
Expand **Goal Node G**

Search Stops.

Search Table

	Step	Expanded Node	Cost	OPEN List
1	S		0	A(2), B(5)
2	A		2	B(5), C(6), G(9)
3	B		5	C(6), G(9)
4	C		6	G(9)
5	G		9	Goal Reached

Search Tree



Pseudocode

START

Insert Start node into Priority Queue

WHILE Queue is not Empty

```

Remove node with minimum cost

IF Goal reached

Return Solution

Expand current node

Insert successors with cumulative cost

END WHILE

STOP

```

4. Accuracy of Calculation (10 Marks)

Path Cost Calculation

Path	Cost
$S \rightarrow A \rightarrow G$	$2 + 7 = 9$
$S \rightarrow A \rightarrow C \rightarrow G$	$2 + 4 + 3 = 9$
$S \rightarrow B \rightarrow C \rightarrow G$	$5 + 2 + 3 = 10$

Least Cost

Minimum Cost = **9**

Optimal Path(s):

- $S \rightarrow A \rightarrow G$
- $S \rightarrow A \rightarrow C \rightarrow G$

Both have the same total cost (**9**).

5. Interpretation of Results (7.5 Marks)

Uniform Cost Search successfully identifies the **least-cost route** by expanding the node with the smallest cumulative cost at each step. This ensures that the first goal node reached is the optimal solution. In logistics, this minimizes transportation cost, fuel consumption, and travel time. UCS is therefore widely used in **delivery networks, GPS navigation, robotics, and route planning**, where finding the cheapest path is essential.